$A = \{a^{3^n} \mid n \geq 0\}$ is not CF.

Proof.
Assume A is CF w/ p.l. k.

Let $z = a^{3^k}$.

Demon picks $uvwxy = z$ s.t.

$|vx| > 0$, $|vwx| \leq k$.

$vx = a^m$ for some $0 < m \leq k$. [why?]

$uv^2wx^2y = a^{3^k + m} = s$. ← just giving a name to uv^2wx^2y

$$\begin{aligned} |s| &= 3^k + m \\ &\leq 3^k + k \\ &< 3^k + 2 \cdot 3^k \\ &= 3^{k+1} \end{aligned}$$

$\therefore s \notin A$, so A is not CF. □

Theorem If L_1 is CF and L_2 is regular, then $L_1 \cap L_2$ is CF.

Proof idea: Nonterminals of the form $A_{q,r}$ derive strings x s.t. $\hat{\delta}(q,x) = r$ and $A \overset{*}{\Rightarrow} x$ in the CFG for L_1.

Proof: Let $G = (N, \Sigma, P, S)$ be a CFG in CNF for L_1.
Let $M = (Q, \Sigma, \delta, q_0, F)$ be a DFA for L_2.
We will build a CFG $G' = (N', \Sigma, P', S')$ for $L_1 \cap L_2$.

$N' = \{A_{q,r} \mid A \in N,\ q, r \in Q\} \cup \{S'\}$

For each rule $A \to BC \in P$ add the $|Q|^3$ rules $A_{q,t} \to B_{q,s} C_{s,r}$ for $q, r, s \in Q$ to P'.

For each rule $A \to t \in P$, add the rules: $A_{q,r} \to t$ for all $q, r \in Q$ s.t. $\delta(q,t) = r$ to P'.

If $S \to \varepsilon \in P$ and $q_0 \in F$, add the rule $S' \to \varepsilon$ to P'.

$\forall f \in F$, add the rule $S' \to S_{q_0,f}$ to P'.

Lemma: For all $A_{q,r} \in N'$ and $x \in \Sigma^* - \{\varepsilon\}$, $A_{q,r} \overset{*}{\Rightarrow} x$ iff $A \overset{*}{\Rightarrow} x$ and $\hat{\delta}(q,x) = r$.

To see that $L(G') = L_1 \cap L_2$, consider 2 cases.

Case 1. $\varepsilon \in L(G') \iff S' \Rightarrow \varepsilon \iff S \Rightarrow \varepsilon$ and $q_0 \in F$
$\iff \varepsilon \in L(G)$ and $\varepsilon \in L(M)$
$\iff \varepsilon \in L_1 \cap L_2$.

Case 2. $x \neq \varepsilon$.

$x \in L(G') \iff S \Rightarrow S_{q_0,f} \overset{*}{\Rightarrow} x$ for some $f \in F$.
$\iff S \overset{*}{\Rightarrow} x$ and $\hat{\delta}(q_0, x) = f \in F$
$\iff x \in L(G)$ and $x \in L(M)$
$\iff x \in L_1 \cap L_2$. □

Proof of lemma by ~~Strong~~ induction on the ~~length of derivation~~ length of x.
Base case $x = t$ for $t \in \Sigma$. $A_{q,r} \Rightarrow t \iff A \Rightarrow t$ and $\delta(q,t) = r$.
Inductive case $|x| > 1$.
$A_{q,r} \Rightarrow B_{q,s} C_{s,r} \overset{*}{\Rightarrow} x$
Apply the I.H.

$B = \{x \in \{a,b,c\}^* \mid \#_a(x) = \#_b(x) = \#_c(x)\}$.
$B \cap L(a^*b^*c^*) = \{a^n b^n c^n \mid n \geq 0\}$ which is not CF
so B is not CF.